

Frobenius 2

Arham Lodha

February 13, 2024

Problem 1. Show that P is diagonalizable, with eigenvalues 0 and 1. Show that $\text{tr}(P) = \dim(P(V))$ in F . Show that if F has characteristic zero, then $P = 0$ iff $\text{tr}(P) = 0$. Explain why such operators are called "projection operators"

Claim 1a. $+ : \ker P \oplus P_*(W) \rightarrow W$ is a isomorphism.

Proof. $\forall v \in \ker P \cap P_*(W)$, since $v \in P_*(V)$, $\exists w \in W \ni Pw = v$. Thus $v = P(w) = P^2(w) = P(v) = 0$, thus $v = 0$. Thus $\ker P \cap P_*(W) = 0$. Thus $+$ is a injective linear map.

By rank-nullity, $\dim \ker P + \dim P_*(W) = \dim W$, thus $+$ is a surjective linear map and thus is a isomorphism. $\square$

Claim 1b. P is diagonalizable and $W_1(P) = P_*(W)$ and $W_0(P) = \ker P$;

Proof. Since $P^2 = P$, P satisfies the polynomial $f(x) = x^2 - x$ which has distinct roots of 0 and 1.

$\forall v \in P_*(W)$, $\exists w \in W \ni Pw = v \implies P^2w = Pv = Pw = v \implies Pv = v$, thus $P_*(W) \subset W_1(P)$. $\forall v \in W_1(P)$, $Pv = v \in P_*(W)$. Thus $W_1(P) \subset P_*(W)$. Thus $P_*(W) = W_1(P)$.

$\forall v \in \ker P$, $Pv = 0 \implies v \in W_0(P)$. $\forall v \in W_0(P)$, $Pv = 0v = 0 \implies v \in \ker P$. Thus $\ker P \subseteq W_0(P)$ and $W_0(P) \subseteq \ker P$. Thus $\ker P = W_0(P)$.

Thus by claim 2a, $W_0(P) \oplus W_1(P) \cong V$. By linear algebra bootcamp, P is diagonalizable. $\square$

Problem 1c. Explain why such a P is called a "projection operator". What is it projection on?

Solution. P is called a projection operator because $P|_{P_*(W)} = \text{Id}_{P_*(W)}$ and it takes everything to the image. $\blacksquare$

Claim 1d. Show $\text{Tr}(P) = \dim(P(V)) * 1 \in \mathbb{F}$

Proof. The trace is the sum of the eignvalues times the multiplicity of the eigenvalue of the matrix. Since P is diagonalizable with eigenvalues 0 and 1, where the number of 0 eigenvectors is $\dim \ker V$ and the number of 1 eigenvectors is $\dim P(W)$. Thus $\text{tr}(P) = 1(\dim P(W)) = \dim P(W)$ $\square$

Problem 2. Let V be a finite dim rep of finite G . $F = \text{field}$ and $|G|1$ is non-zero in F . Write $g : V \rightarrow V$ for the corresponding linear transformation (so as usual, we are leaving the name of the rep, the homo $G \rightarrow GL(V)$, out of the notation). let P be $1/|G| \sum_{g \in G} g$, an element of $\text{Hom}_C(V, V)$. show that $gP = Pg = P$ for all g in G , that $P^2 = P$, and $P(V) = V^G := \{v \in V | gv = v \forall g \in G\}$. Conclude in particular that $1/|G| \sum_{g \in G} \text{tr}(g) = \dim V^G$. In particular it is an integer.

Proof. $\forall h \in G, Ph = (\frac{1}{|G|} \sum_{g \in G} g)h = \frac{1}{|G|} \sum_{g \in G} gh$. Multiplying by h simply reindexes the elements of G , because $\forall g \in G, g = gh^{-1}h = g'h$ where $g' \in G$. Thus $Ph = \frac{1}{|G|} \sum_{g \in G} gh = \frac{1}{|G|} \sum_{g' \in G} g' = P$. Similarly, $hP = \frac{1}{|G|} \sum_{g \in G} hg$. Multiplying by h again simply reindexes the elements of G , because $\forall g \in G \exists g' = h^{-1}g \in G \ni g = hg'$. Thus $hP = \frac{1}{|G|} \sum_{g \in G} hg = \frac{1}{|G|} \sum_{g' \in G} g' = P$. Thus $hP = P = Ph$.

$P^2 = (\frac{1}{|G|} \sum_{g \in G} g)P = \frac{1}{|G|} \sum_{g \in G} gP$. Since $gP = P$, $P^2 = \frac{1}{|G|} \sum_{g \in G} P = |G|/|G|P = P$
 $\forall v \in P_*(V), \exists w \in V \ni Pw = v, \forall g \in G$, by problem 1 $gP = P$. $gv = gPw = Pw = v$. Thus $v \in V^G$. Thus $P_*(V) \subset V^G$.

Thus $\forall v \in V^G, Pv = (\frac{1}{|G|} \sum_{g \in G} g)v = (\frac{1}{|G|} \sum_{g \in G} gv) = (\frac{1}{|G|} \sum_{g \in G} v) = \frac{1}{|G|}|G|v = v \in P_*(V)$. Thus $V^G \subseteq P_*(V)$. Thus $V^G = P_*(V)$

By problem 1, $tr(P) = \dim P_*(V) = \dim V^G$ since $P_*(V) = V^G$. $tr(P) = tr(\frac{1}{|G|} \sum_{g \in G} g) = 1/|G| \sum_{g \in G} tr(g)$ since the trace is linear with regards to addition. Thus $1/|G| \sum_{g \in G} tr(g) = \dim V^G$ $\square$

Problem 3. Let V, W be two finite dim reps of G . We know from the previous homework that G now acts on $H := Hom_F(V, W)$. Show H^G are the set of linear maps from V to W that commute with G . Show (when F is a subfield of $\mathbb{C}$) that $\langle tr_V, tr_W \rangle = tr(P)1 \in F$ where P is $1/|G| \sum_{g \in G} [g : H \rightarrow H]$ (where $\langle \cdot, \cdot \rangle$ is the inner product from the previous homework). Conclude, when F is a subfield of $\mathbb{C}$, that this inner product is an integer, the dimension of $Hom_G(V, W)$.

Solution. Let $f_v : G \rightarrow GL(V), f_w : G \rightarrow GL(W)$ be representation. By definition, $H^G = \{h \in H : gh = h\forall g \in G\}$. In otherwords, $\forall g \in G, f_w(g)^{-1} \circ h \circ f_v(g) = h$, thus $h \circ f_v(g) = h \circ f_w(g)$, thus h commutes with G .

Define a representation, $\gamma : G \rightarrow H \ni g \rightarrow L_{f_v(g)} \circ R_{f_w(g)}$. Thus $\langle tr_v, tr_w \rangle = \frac{1}{|G|} \sum_{g \in G} tr_v(g)tr_w(g)$. By the previous homework, $tr_v(g)tr_w(g) = tr_{L_{f_v(g)} \circ R_{f_w(g)}} = tr_H(g)$. Thus $\langle tr_v, tr_w \rangle = \frac{1}{|G|} \sum_{g \in G} tr_H(g) = tr(P)$.

When F is a subfield of $\mathbb{C}$, by the previous problem $tr(P) = \dim H^G$ which is integer. Thus $\langle tr_v, tr_w \rangle = \dim H^G$ $\blacksquare$

Problem 4. Notation as in 3), take f in $Hom_G(V, W)$ show that $ker(f)$ and $im(f)$ are G -invariant. Prove Schur's lemma: that if V and W are irreducible reps, then $Hom_G(V, W) = 0$ unless V and W are iso. Show further that $Hom_G(V, V)$ is a division ring (ring where every non-zero element is invertible), containing a natural copy of F , which is equality in case F is algebraically closed (every polynomial over F has a root, as holds e.g. for $\mathbb{C}$). Finally, prove the orthogonality of irreducible traces, ie. that the functions tr_W over irreducible reps W of G are an orthogonal set of vectors in $Hom_{sets}(G/G, \mathbb{C})$, for the F valued inner product $\langle X, Y \rangle := 1/|G| \sum_g X(g)Y(g^{-1})$. Show this inner product is non-degenerate on $Hom_{sets}(G/G, \mathbb{C})$. Conclude in particular that there are at most $|G/G|$ iso classes of irreducible reps. When F is a subfield of $\mathbb{C}$, show that traces of irreps have positive self inner product, equal to 1 if $F = \mathbb{C}$.

Proof. $\forall f \in Hom_G(V, W), \forall v \in ker(f), \forall g \in G, f(gv) = gf(v) = 0$ since f commutes with g . Thus $gv \in ker f$. Thus $ker f$ is G -invariant.

$\forall f \in Hom_G(V, W), \forall v \in V, gf(v) = f(gv) \in f_*(V)$. Thus $f_*(V)$ is G -invariant.

$\forall f \in Hom_G(V, W)$, if $f_*(V) = 0$, then $Hom_G(V, W) = 0$ and $ker f = V$, thus $V \not\cong W$. Thus WLOG $f_*(V) \neq 0$. By the above, $f_*(V)$ is G -invariant. Since W is irreducible, by mashke's theorem and previously proven it doesn't have any G -invariant subspaces except itself and 0 subspace. Since

$f_*(V)$ is G -invariant and $f_*(V) \neq 0$, $f_*(V) = W$. Similarly, $\ker f$ is a G invariant subspace in V . Again by mashke's theorem, since V is irreducible it has no G -invariant subspaces other than itself and the trivial zero subspace however since $f_*(V) \neq 0 \implies \dim \ker f < \dim V$ thus $\ker f = 0$. Thus f is a bijection and isomorphism of G -reps. Thus this means $\dim Hom_G(V, W) \neq 0 \iff V \cong W$ as G -reps.

$\forall f \in Hom_G(V, V)$, everything in f is either 0 or a isomorphism or the above. Thus f is either 0 or invertible thus $Hom_G(V, V)$ is a division ring.

$\forall f \in Hom_G(V, V)$, choose a eigenvalue λ of f . $S = f - \lambda I$ be a linear map from $V \rightarrow V$. The $\ker S \neq 0$ because it contains a eigenvector of f . Thus S is not a isomorphism. By the above, $S = 0 \implies f - \lambda I = 0 \implies f = \lambda I$. Thus $F \subset Hom_G(V, V)$ naturally and if λ is nonzero, f is invertible. Thus $Hom_G(V, V)$ is a division ring. If F is algebraically closed, all eigenvalues of matrices can be found and thus $F = Hom_G(V, V)$. $\square$

Claim 4b. *Prove the orthogonality of irreducible traces, ie. that the functions tr_W over irreducible reps W of G are an orthogonal set of vectors in $Hom_{sets}(G/G, C)$, for the F valued inner product $\langle X, Y \rangle := 1/|G| \sum_g X(g)Y(g^{-1})$.*

Proof. Let V, W be irreducible representations of G . If $V \cong W$, then $Hom_G(V, W)$ consists of scalar matrices as shown above thus $\dim Hom_G(V, W) = 1$. If $V \not\cong W$, then $Hom_G(V, W) = 0$ as proven above thus $\dim Hom_G(V, W) = 0$. From 3, we know $\langle Tr_V, Tr_W \rangle = \dim H^G = Hom_G(V, W)$ which equals 1 or 0 depending on if $V \cong W$ as shown above. Thus Tr_V is orthogonal to Tr_W if $V \not\cong W$ and $\langle Tr_V, Tr_W \rangle = 1$ if and only if they are isomorphic. $\square$

Claim 4c. *Show this inner product is non-degenerate on $Hom_{sets}(G/G, C)$.*

Proof. It suffices to prove that if $\forall \alpha \in Hom_{sets}(G/G, \mathbb{C}) \ni \alpha \neq 0, \exists \beta \in \langle \alpha, \beta \rangle \neq 0$.

$\forall \alpha \in Hom_{sets}(G/G, \mathbb{C}) \ni \alpha \neq 0$. Thus $\langle \alpha, \alpha \rangle = \frac{1}{|G|} \sum_{g \in G} \alpha(g)\overline{\alpha(g)} = \frac{1}{|G|} \sum_{g \in G} |\alpha(g)|$. Since $\alpha \neq 0$, thus *exists* $g \in G \ni \alpha(g) \neq 0 \implies |\alpha(g)| \neq 0$. Furthermore, since modulus is positive for all nonzero values and zero for zero, $\frac{1}{|G|} \sum_{g \in G} |\alpha(g)| > 0$. Thus $\langle \alpha(g), \alpha(g) \rangle \neq 0$. Thus $\langle \rangle$ is nondegenerate. $\square$

Problem 4d. *Conclude in particular that there are at most $|G/G|$ iso classes of irreducible reps. When F is a subfield of C , show that traces of irreps have positive self inner product, equal to 1 if $F = C$.*

Proof. The set of traces of irreducibles are linearly independent on the account of being orthogonal and since $\dim Hom_{sets}(G/G, F) = |G/G|$. The dimension of the span of the traces cannot be greater than G/G , thus the number of traces of irreducibles is at most G / G .

Suppose $F \leq \mathbb{C}$. Let V be a irreducible representation of G . The $\langle tr_V, tr_V \rangle = \sum_{g \in G} tr_V(g)\overline{tr_V(g)} = \dim Hom_G(V, V) = 1$, by the above since $Hom_G(V, V)$ consists of scalar matrices. $\square$

Problem 5. *Prove, for F a subfield of C , that two finite dim F reps of finite G are iso iff they have the same trace - hint write a rep V as a direct sum of irreducibles, and take the inner product of tr_V with tr_{V_i} for each irreducible V_i .*

Proof. Suppose V, W are finite representations of G . Let $\{X_i\}$ be the set of all irreducible representations of G . Let there be n irreducibles representations of G . By representations 1, $V \cong \bigoplus_{i=1}^n X_i^{m_i}$ and $W \cong \bigoplus_{i=1}^n X_i^{k_i}$.

($\implies$) Suppose $V \cong W$, Where m_i are the multiplicities of the irreducibles, or the number of times they would have been in the direct sum, in the vectorspace, Since $V \cong W \implies \bigoplus_{i=1}^n X_i^{m_i} \cong$

$\bigoplus_{i=1}^n X_i^{k_i} \implies m_i = k_i$. Since X_i are G -invariant, $\forall g$, g is block diagonal on V and W . Thus $tr_V = \sum_{i=1}^n m_i tr_{X_i} = \sum_{i=1}^n k_i tr_{X_i} = tr_W$. Thus the traces are equal.

($\Leftarrow$), Suppose $tr_V = tr_W$, where V and W are finite dimensional vector spaces. By the above, $tr_V = \sum_{i=1}^n m_i tr_{X_i}$ and $tr_W = \sum_{i=1}^n k_i tr_{X_i}$. $\langle tr_V, tr_{X_i} \rangle = m_i$ since $\langle tr_{X_i}, tr_{X_j} \rangle = \delta(i, j)$, by Schur's lemma. Thus $m_i = \langle tr_V, tr_{X_i} \rangle = \langle tr_W, tr_{X_i} \rangle = k_i$. Thus $V \cong \bigoplus_{i=1}^n X_i^{m_i} = \bigoplus_{i=1}^n X_i^{k_i} \cong W$. Thus $V \cong W$. $\square$

Problem 6. Let $f : G \rightarrow F$ be a class function (constant on conjugacy classes). And assume $\text{char}(F)$ does not divide $|G|$. For a finite dim G -rep V let $P := 1/|G| \sum_{g \in G} f(g) g$ in $\text{Hom}_F(V, V)$. Show that $gP = Pg$. Now suppose $F = \mathbb{C}$, and f is orthogonal to tr_V for each irreducible rep V . Show $f = 0$. Hint: Show that $P = 0$ (for any V). For this show its enough to prove the case when V is irreducible. Using Schur's lemma, show then its enough to prove $tr_P = 0$, and get this from the orthogonality. So for all V (irreducible or not) $P : V \rightarrow V = 0$, use this to prove f is zero. Conclude, when $F = \mathbb{C}$, that the traces of irreps are a basis of class function. Conclude we have $|G/G|$ irreducible reps. We'll show later this is true for any algebraically closed F (so long as $\text{char}(F)$ does not divide $|G|$).

Proof. Let $V = F^G$ be a G -rep. In this representation, $G \subset S_{|G|}$ and are the permutation matrices and thus their columns are the standard basis in $F^{|G|}$. $\forall M, N \in G \ni M \neq N$, there is some $i \in \mathbb{N}_{|G|}$ $Me_i \neq Ne_i$, thus none of the permutation matrices all fix an individual standard basis. Thus suppose $\sum_{i=1}^{|G|} \alpha M_i = 0 \implies (\sum_{i=1}^{|G|} \alpha M_i)e_j = 0$ for all j . $\implies \sum_{i=1}^{|G|} \alpha M_i e_j$ however since none of the standard basis are preserved by all the matrices e_j will be sent to different basis e for at least 1 matrix thus $0 \leq \alpha_k e = \sum_{i=1}^{|G|} \alpha M_i e_j = 0$. Thus α must be 0. Thus the set of permutations are linearly independent thus G is linearly independent.

Let $f \in G \rightarrow F$ be a class function. Let $P = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} g \in \text{Hom}_{V,V}$, thus g in this context could be the element of G or corresponding representation of G in $GL(V)$. $hPh^{-1} = h \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} gh^{-1} = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} hgh^{-1} = \frac{1}{|G|} \sum_{g \in G} \overline{f(hgh^{-1})} hgh^{-1}$, because f is invariant under conjugation. Furthermore, conjugation by fixed elements rotates the elements in their orbits and thus reindexes the elements of G . Thus, $hPh^{-1} = \frac{1}{|G|} \sum_{g \in G} \overline{f(hgh^{-1})} hgh^{-1} = \frac{1}{|G|} \sum_{g' \in G} \overline{f(g')} g' = P$. Since $V = \bigoplus X_i^{m_i}$ where X_i is G -invariant, $\forall g \in G$, g is block diagonal. Thus P is block diagonal. Moreover since P commutes with G , on X_i , it is a scalar matrix by Schur's Lemma as $P|_{X_i} \in \text{Hom}_G(X_i, X_i)$. Thus we can consider each irreducible block separately. $tr_{X_i}(P|_{X_i}) = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} tr_{X_i} g = \langle tr_{X_i}, f \rangle = 0$, by assumption. The only scalar matrix with trace 0 is the 0 matrix. Thus $\forall X_i, P|_{X_i} = 0 \implies P = 0$ since P is block diagonal. Thus $P = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} g = 0$, since $\forall g \in G$ are linearly independent, $\overline{f(g)} = 0 \implies f(g) = 0 \implies f = 0$. Thus when $F = \mathbb{C}$, the traces of irreducibles form a basis of class functions. Since every isomorphic representation has a equal trace, and non isomorphic representations have nonequal traces and there are $|G/G|$ traces for irreducibles, since the traces form a basis of $\text{Hom}_{sets}(G/G, \mathbb{C})$, there are also $|G/G|$ irreducible G -reps. $\square$